Rethinking Content Moderation via Severity-Aware Ranking

Yang Ou[*], **Jielei Zhang**[*†], **Yu Dai, Zian Yang, Run Yang,**
Xiang Li, Weihang Wang, Xueqian Zhao,
Peiyi Li, Longwen Gao

Bilibili Inc.
{oy735899115, yctmzjl}@gmail.com
{daiyu03, yangzian, yangrun, lixiang20,
wangweihang01, zhaoxueqian, lipeiyi, gaolongwen}@bilibili.com

Abstract

Online video content moderation is critical for protecting minors. Violations are increasingly implicit and context-dependent: a swimwear is innocuous in a vlog but high-risk with suggestive posing. Yet, traditional binary systems fundamentally fail to capture this continuous severity spectrum, directly leading to both excessive false positives for context-heavy samples and missing deeply implicit violations. We introduce KUROKO, a content moderation system that acts as a behind-the-scenes assistant, driving a paradigm shift in content moderation from binary classification to severity-aware ranking. Our contributions are threefold: (1) Human-Centric Severity Ranking: KUROKO shifts from rigid binary divides to a continuous risk spectrum, producing nuanced, expert-level rationales closely aligning with human judgment. (2) Verifiable Alignment and Interpretability: By integrating a severity-aware data engine and a verifiable reinforcement learning approach, we replace black-box decisions with transparent, actionable diagnostics. (3) Cost-Efficient Deployment: We engineer a coarse-to-fine inference architecture equipped with a lightweight recall classifier, enabling compact models to rival proprietary VLMs at a fraction of the computational cost. Extensive offline and online A/B tests demonstrate KUROKO's advantages over traditional binary baselines and its competitiveness with leading commercial APIs at a fraction of their scale. Online, KUROKO yields more human-confirmed enforcement actions under an equal human-review budget, while lowering violative exposure and user negative feedback. Source code and dataset are available at: https://bilibili-kuroko.github.io.

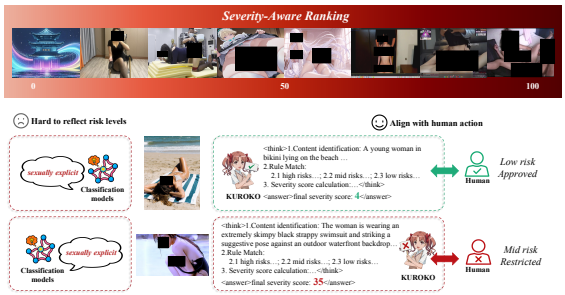


Figure 1: From binary to severity ranking. KUROKO evaluates compositional context for interpretable, severity-aware assessment, precisely distinguishing benign from provocative content.

1 Introduction

Online video sharing platforms (OVSPs), such as YouTube, serve as primary hubs for entertainment and education, making content moderation essential to safeguard users from harmful material (Gillespie, 2018). Conventional moderation systems (Wang et al., 2025b; Yew et al., 2026) formulate this task as binary classification: samples are labeled "violating" or "non-violating" against rigid rules, and models are trained to reproduce these boundaries. However, as illustrated in Figure 1, real-world content exists on a continuous risk spectrum with a vast gray area that defies binary categorization (Kiela et al., 2020). This mismatch leads to both severe over-penalization of benign-but-sensitive content and missed violations that exploit boundary ambiguity. Furthermore, the black-box nature of binary decisions offers no diagnostic insight, leaving developers trapped in directionless trial-and-error (Ribeiro et al., 2016).

Recent efforts (Redmon et al., 2016; Bao et al., 2025; Liu et al., 2023; Li et al., 2023; Chen et al., 2024) leverage Vision-Language Models (VLMs) for richer contextual understanding, yet remain locked in the classification paradigm—mapping complex visual inputs onto discrete binary labels while failing to provide the severity-aware assessment necessary to navigate the gray area.

[*] Equal contribution.
[†] Corresponding author.

To bridge these gaps, we present **KUROKO**, an industrial-scale content moderation system built on a generative severity ranking paradigm, to our knowledge among the first deployed in production at this scale. Our contributions are summarized as follows:

- **Severity Ranking Paradigm.** We formalize a shift in content moderation from rigid binary classification to a continuous severity-aware ranking framework. This paradigm tackles the long-standing "gray area" bottleneck, mitigating over-penalization while substantially narrowing the alignment gap with human judgment.
- **Rubric-Verifiable Reinforcement Learning with SACRM.** We propose a Rubric Reinforcement Learning with Verifiable Rewards (RLVR) approach for severity-aware moderation alignment. At its core is a novel Severity-Aware Constrained Rule-Matching (SACRM) reward—a programmatic rubric enforcing multi-tier precision and hard guardrails—which delivers verifiable alignment signals and interpretable diagnostics where conventional preference rewards fall short.
- **Industrial-Grade Pipeline from Data to Deployment.** We engineer an end-to-end pipeline that operationalizes our method at production scale. Our severity-aware data engine structures expert audit logic into reasoning trajectories, achieving 5.56× the annotation efficiency of manual labeling for SFT and RL. Simultaneously, a coarse-to-fine inference architecture—pairing a lightweight recall classifier with the generative VLM—shields the system from redundant traffic, allowing KUROKO to handle millions of daily requests while rivaling proprietary commercial APIs at a fraction of the computational cost.

2 Related Works

2.1 Content Moderation in Video Platforms

While human moderation was foundational, its scalability and welfare challenges—driven by rising user generated content volumes, reviewer subjectivity (Lyons et al., 2021), and psychological harm (Gonzalez and Matias, 2025; Roberts, 2014)—have accelerated the transition toward automation (Lykouris and Weng, 2025). However, early automated systems rely heavily on Classification-based Models (CBMs) (Won et al., 2017; Khan et al., 2024; Phan et al., 2022; de Freitas et al., 2019), which detect only predefined patterns. Consequently, they lack the granularity and adaptability required for evolving content risks, yielding low interpretability and a narrow scope of applicability. Although ordinal classification (Niu et al., 2016; Cao et al., 2020) addresses this by modeling ordered risk levels rather than flat labels, its $K-1$ cumulative thresholds still collapse to one discrete tier at inference, leaving samples within a tier mutually interchangeable and offering no interpretable rationale—bounding both granularity and diagnostic utility for iterative refinement.

2.2 VLMs for Content Moderation

While promising CBM successors, open-source VLMs lack domain moderation knowledge due to prohibitive curation costs (AlDahoul et al., 2024; Wang et al., 2025c), and leveraging synthetic data risks overfitting (Wu et al., 2025; Cui et al., 2025). Beyond data, current VLM-based methods—such as Chain-of-Thought (CoT) (Lu et al., 2025), multi-dimensional frameworks (Bao et al., 2025), and RL alignment (Li et al., 2026)—remain trapped in a classification paradigm. They struggle with evolving borderline cases and incur severe computational overhead compared to lightweight CBMs. To mitigate these inference costs, various efficient reasoning strategies have emerged, encompassing CoT compression (Kang et al., 2025), token pruning (Xia et al., 2025), prompt guidance (Xu et al., 2025; Lee et al., 2025; Han et al., 2025), distillation (Li et al., 2024; Liao et al., 2025; Li et al., 2025), latent reasoning (Hao et al., 2024; Shen et al., 2025), and dynamic adaptive CoT (Lou et al., 2025; Zhang et al., 2025).

3 Methodology

3.1 Problem Formulation

Traditional CBMs treat moderation as a binary decision over a rigid action space $\mathcal{A}_0 = \{\text{Accept}, \text{Reject}\}$, collapsing the risk spectrum so that high recall floods reviewers with false positives while high precision lets harmful content reach users. Mildly suggestive videos and explicitly exploitative material receive identical binary labels, ignoring the *compositional semantics* of

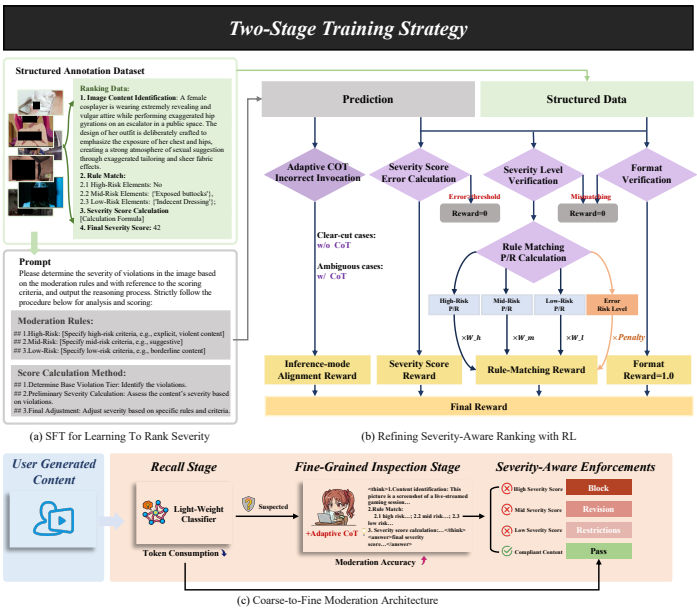


Figure 2: **KUROKO pipeline.** (a) *SFT* leverages adaptive reasoning for policy grounding; (b) *RL via Rubric RLVR* optimizes tier-wise alignment; (c) *Coarse-to-Fine Deployment* filters and routes risky samples.

moderation rules, where identical visuals yield divergent severity by context (Figure 1). In practice, OVSPs already operate over a graded action space $\mathcal{A}^+ = \{\text{Accept}, \text{Restrict}, \text{Revision}, \text{Reject}\}$ that matches enforcement to assessed risk, motivating an ideal moderator $f_{\text{ideal}} : \mathcal{C} \times \mathcal{P} \to \mathcal{A}^+$ (where $\mathcal{C}$ denotes the user content space and $\mathcal{P}$ the policy configurations) that should possess two capabilities: (i) a fine-grained rule-grounding function $g : \mathcal{C} \times \mathcal{R}_p \to [0, 1]$ that identifies which rules from a finite set $\mathcal{R}_p$ are violated, and (ii) a severity aggregation that mirrors human reviewers in jointly weighing the violated rules and their respective risk levels to produce a final score $s_p : \mathcal{C} \to \mathbb{R}$ aligned with human judgment:

$$s_p(c) = \phi\left(\{d_p(r) \cdot g(c, r) \mid r \in \mathcal{R}_p\}\right), \qquad (1)$$

where $d_p(r)$ encodes per-rule risk weights and ϕ aggregates rule-level signals into a scalar severity. The resulting score is then mapped monotonically via $h_p : \mathbb{R} \to \mathcal{A}^+$ to graded actions, completing f_{ideal}.

3.2 Policy-Grounded Alignment and Adaptive Reasoning

Open-source VLMs lack domain-specific moderation knowledge, and their safety alignment often induces over-refusals on sensitive inputs. To address this, our severity-aware data engine (Figure 3, Appendix A) structures expert audit logic into fine-grained trajectories. Crucially, rather than forcing the model to blindly memorize the combinatorial space of rule violations, we explicitly train it to replicate the systematic two-step workflow of human reviewers: (1) identifying specific policy violations (g), and (2) aggregating the severity based on predefined logic (ϕ). The exact weights (d_p) and aggregation mechanics of our production system are redacted to protect commercial enforcement standards from circumvention; Appendix C gives an executable public proxy that reproduces every reported metric without them, and h_p is likewise a configurable policy mapping rather than a fixed score-to-action lookup. By performing SFT on these trajectories, the model internalizes the rule-based rationale (g, ϕ) rather than memorizing fixed point assignments, so that the weight schedule d_p can be re-parameterized at deployment without retraining.

Since applying full CoT reasoning to every item is computationally prohibitive in production, we adopt an adaptive reasoning strategy (Lou et al., 2025). A lightweight complexity router (Appendix G.2) partitions training data into CoT-needed subsets (ambiguous gray-area cases) and CoT-unneeded subsets (obvious violations or clearly safe content). Fine-tuning on this mixed-granularity dataset enables the model to internalize both detailed reasoning for hard samples and direct prediction for easy ones, minimizing average latency without sacrificing depth on difficult cases. The model is optimized via standard next-

token prediction loss:

$$\mathcal{L}_{\text{SFT}} = -\sum_{t=1}^{T} \log P\left(y_t \mid y_{<t}, x; \theta\right). \quad (2)$$

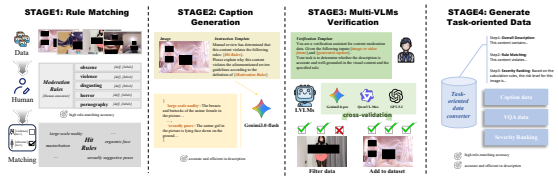


Figure 3: Overview of our severity-aware data engine.

3.3 Severity Alignment via Rubric RLVR

While SFT establishes baseline severity-ranking capability, the resulting model exhibits two residual issues: systematic errors in risk-level stratification, and output instability when toggling between direct and reasoning modes. To enforce rigorous compliance, we introduce Rubric RLVR, powered by a programmatic SACRM reward scheme to stabilize training and align optimization with operational risk priorities (see Figure 2b).

Reliable severity ranking requires precise localization of policy rules in visual evidence and correct association with designated risk levels. We optimize per-level precision $\mathcal{R}_{\text{prec}}^{(k)}$ and recall $\mathcal{R}_{\text{rec}}^{(k)}$ across risk tiers k, weighted by importance w_k:

$$\mathcal{R}_{\text{rule}} = \sum_{k=1}^{K} w_k \left(\alpha \mathcal{R}_{\text{prec}}^{(k)} + \beta \mathcal{R}_{\text{rec}}^{(k)}\right) \quad (3)$$

where $\sum_{k=1}^{K} w_k = 1$, and α, β trade off precision and recall.

To facilitate stable policy updates under structured rewards, the SACRM rubric incorporates **four specialized components** targeting distinct operational failure modes:

(1) Soft severity-tier penalty. Predictions at an incorrect risk level receive a $0.5\times$ discount on $\mathcal{R}_{\text{rule}}$, encouraging accurate stratification while preserving learning gradients.

(2) Hard deviation guardrail. If the predicted score deviates severely from the ground truth ($|s_{\text{pred}} - s_{\text{gt}}| > \tau$), the sample receives zero total reward, preventing catastrophic mis-rankings from dominating optimization.

(3) Inference-mode alignment reward $\mathcal{R}_{\text{align}}$. This reward penalizes inconsistencies between the model's actual inference behavior and the router's intent, ensuring correct utilization of direct versus reasoning modes.

(4) Format consistency reward $\mathcal{R}_{\text{fmt}}$. This reward penalizes structural deviations from the prescribed output template required by downstream production pipelines.

The final composite reward is formalized as:

$$\mathcal{R}_{\text{tot}} = \begin{cases} 0, & |s_{\text{gt}} - s_{\text{pred}}| > \tau \\ \sum_i \gamma_i \mathcal{R}_i, & \text{otherwise} \end{cases} \quad (4)$$

where $(\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3) = (\delta\,\mathcal{R}_{\text{rule}}, \mathcal{R}_{\text{align}}, \mathcal{R}_{\text{fmt}})$ with corresponding weights $\gamma_1, \gamma_2, \gamma_3$, and $\delta = 0.5$ if the predicted risk tier differs from the ground truth (else $\delta = 1$).

3.4 Cost-Efficient Coarse-to-Fine Deployment

To mitigate the computational overhead of brute-force VLM deployment under massive traffic, KUROKO adopts a hierarchical coarse-to-fine architecture. In the coarse-grained triage filter stage, a lightweight, high-recall model rapidly screens all incoming traffic, immediately clearing obviously compliant items, shielding the VLM from redundant load. In the fine-grained inspection stage, the VLM executes severity-ranking CoT exclusively on the recalled subset. Crucially, the model's internalized adaptive reasoning automatically toggles between *direct mode* (bypassing text generation for easy samples) and *reasoning mode*. Because cleared items never reach the VLM, this stage's recall bounds end-to-end safety; we report its false-negative rate in Sec. 4.5.

4 Evaluation

4.1 Experimental Setup

4.1.1 Dataset and Benchmark

Via our data engine (Appendix A), we curate large-scale datasets across five domains using verified production data. For efficiency, clear-cut samples use empty reasoning (`<think></think>`), while ambiguous cases receive full CoT rationales. We evaluate on our introduced Fine-Grained Content Moderation Benchmark (FG-CMB). Its four severity tiers (*Safe, Low-Risk, Medium-Risk, High-Risk*) directly assess ranking performance and im-

	Training			FG-CMB		
Category	**H**	**M**	**L**	**H**	**M**	**L**
Pornography	2,311	0	0	351	0	0
Obscene	1,910	5,025	6,038	254	648	647
Violence	721	150	5	609	117	33
Horror	284	6	0	139	19	5
Disgusting	107	1,422	0	60	1,070	2
Normal		70,000			9,978	

Table 1: Category-wise distribution of training data and FG-CMB across severity tiers. Category rows are not mutually exclusive: a frame matching several categories counts in each. FG-CMB is hard-case-enriched rather than a random production draw, over-sampling historical false negatives, subtle violations and small targets.

plicit/explicit violation discrimination (Table 1, Appendix A.3).

4.1.2 Evaluation Metrics

We evaluate KUROKO across three dimensions to reflect both safety screening accuracy and severity discrimination: (1) **Binary Safety Metrics**: Precision, Recall, and F1-score for Safe vs. Unsafe classification. (2) **Severity-Aware Metrics**: Per-tier (High, Medium, Low) evaluation of stratification reliability near policy boundaries. Per-severity results report *stratified binary detection* (each level against Safe samples); the matched comparison instead scores *exact severity assignment* and adds **Tier MAE** (mean absolute tier distance), **QWK** (quadratic-weighted κ), **Spearman** against the rule-derived reference score, and **Intra-tier ranking** (pairwise ordering accuracy *within* each true level, ties 0.5). Intra-tier reference order follows deterministically from unanimous three-annotator rule labels via the fixed aggregator, never a model output. (3) **Online A/B Metrics**: human-final Reject and Restrict rates at an equal review budget, plus Violative View Rate and User Dislike Rate, to verify proportional enforcement without relying on flag volume alone.

4.1.3 Baselines

(1) Classification Baselines: We compare against Falconsai NSFW[1] and an in-house binary classifier built on Qwen3-VL-4B-Instruct with identical data. We further fine-tune a CORAL ordinal head (Cao et al., 2020) on the Falconsai NSFW CNN with partitioned ordinal labels. Because that backbone is far weaker than ours, it cannot separate paradigm from capacity; we therefore add a **matched ordinal head** on Qwen3-VL-4B-Instruct (Bai et al., 2025) backbone under identical data and step budget, which is the comparison that isolates the ranking paradigm. **(2) Frontier VLMs**: We benchmark against leading commercial APIs (GPT-5.2, Gemini-3.0-Pro, Gemini-3.0-Flash and Qwen-VL-Max) to evaluate performance. **(3) Production System**: We evaluate KUROKO via large-scale A/B tests against the heavily optimized live system, which uses a VLM classifier relying on direct logit scores rather than our generative ranking.

[1] https://huggingface.co/Falconsai/nsfw_image_detection

4.1.4 Implementation Details

All offline experiments use a single node with 4 NVIDIA A100 GPUs. SFT fine-tunes Qwen3-VL-4B-Instruct for 2 epochs via Megatron-LM (Shoeybi et al., 2019) (lr 1×10^{-5}, batch size 256, warmup 0.1). RL employs GSPO (Zheng et al., 2025) via VeRL (Sheng et al., 2025), sampling 8 responses per query with batch size 128. For API-based VLMs, we use a curated prompt strategy (Appendix B) to balance fairness and costs. The online A/B test runs on the live pipeline (20% traffic, five days), deploying a Recall Classifier (1 A10 GPU) followed by the VLM ranking module (2 RTX 4090 GPUs). All four relative changes in Table 5 are significant at $p < .05$ under two-sided two-proportion z-tests.

4.2 Offline Experimental Results

Tables 2 and 3 evaluate KUROKO across risk domains and severities under the stratified binary detection protocol. Despite its 4B size, KUROKO surpasses the larger Qwen-VL-Max across all five domains (e.g., F1 +10.40% in *Pornography*, +12.16% in *Disgusting*) and beats the data-identical Qwen3-VL-4B-FT (Cls), indicating the gains do not come from the training data alone. The advantage peaks in ambiguous tiers: KUROKO's Medium-tier F1 (67.20%) leads every API evaluated here, though Appendix D shows Gemini-3.0-Pro closing much of this gap once all models are given an identical rubric. Holding backbone, data, and step budget fixed, KUROKO-SFT leads the ordinal head Qwen3-VL-4B-FT (Ord) in F1 at every severity level (+7.1, +5.3 and +12.1 pp) and at higher precision throughout, with RL and coarse-to-fine adding a further +3.2 to +4.9 pp. In the Low tier

Method	Pornography			Obscene			Violence			Horror			Disgusting		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
NSFW-Classifier	26.94	56.41	36.47	41.95	41.19	41.57	2.36	1.87	2.09	1.47	5.76	2.34	33.13	24.81	28.37
Gemini-3.0-Pro	33.48	86.32	48.25	55.87	80.89	66.08	32.70	57.49	41.69	11.72	78.42	20.39	52.35	84.14	64.54
GPT-5.2	24.01	83.19	37.27	45.45	81.74	58.42	20.87	78.96	33.01	4.94	77.70	9.29	31.49	89.09	46.53
Gemini-3.0-Flash	30.38	84.05	44.63	53.37	82.17	64.71	23.20	59.37	33.36	6.51	68.34	11.89	31.18	57.65	40.47
Qwen-VL-Max	32.68	66.38	43.80	55.84	64.44	59.83	31.33	61.53	41.52	9.65	71.94	17.02	48.17	81.16	60.46
Qwen3-VL-4B-Instruct	13.31	44.73	20.52	31.02	48.83	37.94	25.44	50.29	33.79	8.50	68.35	15.12	25.51	34.42	29.30
Qwen3-VL-4B-FT (Cls)	23.91	87.22	37.53	48.15	83.58	61.10	19.60	72.16	30.82	4.83	79.53	9.10	34.86	87.44	49.85
KUROKO	42.01	76.35	**54.20**	63.97	69.75	**66.74**	42.63	53.31	**47.38**	14.14	58.99	**22.81**	64.24	83.48	**72.62**

Table 2: Comprehensive evaluation on FG-CMB across five risk domains (values in %), benchmarking KUROKO against various classification baselines, leading commercial VLM APIs, and Qwen-family models.

Method	High			Medium			Low		
	P	R	F1	P	R	F1	P	R	F1
CORAL	23.44	38.75	29.21	32.54	37.41	34.81	27.65	82.38	41.40
Gemini-3.0-Pro	37.22	80.49	50.90	50.81	85.88	63.84	27.48	87.64	41.85
GPT-5.2	23.71	84.66	37.05	34.63	88.38	49.76	16.38	90.88	27.76
Gemini-3.0-Flash	29.55	89.29	44.40	41.31	91.77	56.97	20.01	90.73	32.79
Qwen-VL-Max	35.61	71.05	47.44	48.83	75.10	59.18	26.82	80.22	40.20
Qwen3-VL-4B-Instruct	36.14	52.54	42.83	39.89	37.74	38.79	26.30	56.41	35.87
Qwen3-VL-4B-FT (Cls)	29.66	81.76	43.53	42.80	88.83	57.76	21.93	92.74	35.47
Qwen3-VL-4B-FT (Ord)	29.44	79.13	42.91	43.04	87.77	57.75	20.08	81.14	32.19
KUROKO-SFT	37.66	74.32	49.98	51.74	80.82	63.09	29.56	87.94	44.25
KUROKO	42.71	70.42	**53.17**	57.99	79.88	**67.20**	34.57	85.01	**49.15**

Table 3: Severity-aware evaluation on FG-CMB across High-, Medium-, and Low-severity levels (values in %), contrasting KUROKO with both ordinal and fine-tuned binary classifiers, leading commercial VLM APIs, and Qwen-family models.

Model	Mac.F1	MAE↓	QWK	Spear.	Intra
Qwen3-VL-4B-FT (Ord)	.4361	.5455	.5248	.5371	58.55
KUROKO-SFT	.5557	.3928	.5749	.6075	68.02
KUROKO-RL	**.5662**	**.3443**	**.6107**	**.6376**	**69.67**

Table 4: Matched comparison against an ordinal head on the *same* backbone. Mutually exclusive four-way tier protocol; Mac.F1 averages over all four tiers including Safe, so it is not comparable to the per-tier F1 in Table 3. Intra-tier in %.

Metric	Relative Δ
Human-final Reject Rate (explicit high risk)	**+12.21%**
Human-final Restrict Rate (mid-low boundary)	**+13.28%**
Violative View Rate	**−23.68%**
User Dislike Rate	**−8.88%**

Table 5: Large-scale online A/B tests in production. Both arms are held to the same human-review intake volume, rule pool, moderators and action definitions, so all four rows are measured at an **equal review budget**. Reject/Restrict count human-confirmed decisions, not model flags.

(implicit violations), the binary FT reaches only 35.47% F1 and the matched ordinal head 32.19%, while KUROKO attains 49.15%—a contrast of binary, ordinal and ranking formulations on an identical backbone that further validates the advantage of the ranking formulation. It also mitigates API over-sensitivity: GPT-5.2's Low-tier precision collapses to 16.38%, while KUROKO sustains 34.57%. Sec. 4.3 scores the same matched pair on exact severity assignment and within-tier ordering.

4.3 Does Continuous Ranking Help Beyond Ordinal Classification?

Stratified binary detection cannot see ordered error or within-tier ordering—any prediction above Safe counts alike—so we rescore Qwen3-VL-4B-FT (Ord) and KUROKO on exact severity assignment, where the budget-matched contrast with KUROKO-SFT leaves how severity is *represented* as the only difference. As Table 4 shows, with backbone and data fixed KUROKO-SFT leads on every metric, all scored against

Method	Pornography			Obscene			Violence			Horror			Disgusting		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Severity-aware SFT	41.43	77.78	54.06	62.67	68.79	65.59	34.26	62.10	44.16	10.40	69.06	18.08	51.67	82.46	63.53
+ RL Refinement	41.48	77.78	54.11	63.40	70.80	**66.90**	41.62	54.76	47.29	13.61	60.43	22.22	62.81	83.96	71.86
+ Coarse-to-fine (Full)	42.01	76.35	**54.20**	63.97	69.75	66.74	42.63	53.31	**47.38**	14.14	58.99	**22.81**	64.24	83.48	**72.62**

Table 6: Ablation study evaluating the cumulative impact of each KUROKO component.

Method	Resp. Len.	Skip Rate	VLM Traffic
Full CoT	565	—	100%
Adaptive CoT	316 ($\downarrow$ 44.07%)	48.38%	100%
C2F + AdaCoT	—	—	22.50%

Table 7: Efficiency analysis of KUROKO variants.

the same severity labels: Macro-F1 $+12.0$ pp (95% CI $[+10.5, +13.5]$), Tier MAE -28.0% relative, QWK $+5.0$ pp, intra-tier $+9.5$ pp, all paired-bootstrap CIs excluding zero; RL widens every margin (QWK $+3.6$ pp). The ordinal head's inductive bias works as intended—its QWK trails KUROKO-SFT by only 5.0 pp against 12.0 pp on Macro-F1, so wrong tiers are mostly adjacent—but it cannot express *within-tier* ordering: $K-1$ cumulative thresholds collapse to one discrete tier at inference, leaving samples inside a tier interchangeable. We therefore claim an advantage in ordered error, rank correlation and within-tier resolution rather than uniform superiority over all classification variants—a ranking that holds across all 11 public proxy weights (Appendix C).

4.4 Results of Online A/B Tests in Production

Beyond offline FG-CMB, we further validate KUROKO under live traffic. As shown in Table 5, compared against the heavily optimized production VLM relying on logit thresholds, KUROKO raises the human-final Reject Rate by **+12.21%** for explicit high-risk content and the Restrict Rate by **+13.28%** for implicit mid-to-low risk boundary cases. Because review intake was held equal across arms, these rows separate *better* ranking from merely *more aggressive* flagging: a ranker that simply flagged more would crowd the fixed queue with benign items and thereby *reduce* human-confirmed actions. We observe the reverse: KUROKO enhances finite human review capacity to intercept actual violations early. Two downstream metrics confirm this precision gain without collateral damage: Violative View Rate drops **23.68%** and User Dislike Rate falls **8.88%**, proving severe items and grey-zone violations are intercepted before impacting users at scale. *bounded* budget.

4.5 Ablation Studies

Table 6 traces the cumulative contribution of each component, starting from Severity-aware SFT. RL Refinement via SACRM yields its largest gain on implicit violations (*Disgusting* F1 $+8.33$ pp), where soft penalties amplify supervision on the ambiguous Low/Medium tiers that standard SFT handles poorly. Coarse-to-fine changes F1 only marginally ($+0.1$ to $+0.8$ pp, -0.2 pp on *Obscene*); its contribution is throughput, quantified next. Table 7 confirms KUROKO's efficiency: Adaptive CoT skips reasoning on 48.38% of unambiguous samples, reducing average response length by 44.07%, while C2F offloads 77.50% of traffic. Since anything the first stage clears bypasses VLM inspection, its false negatives bound end-to-end safety: at the production operating point it recalls 98.61% of violating samples. These are *benchmark* rates on the hard-case-enriched FG-CMB, not natural-traffic estimates; Appendix E gives a 5% online audit and the coverage-parity gating that holds this level.

5 Conclusion

We presented KUROKO, an industry-ready moderation framework that shifts the paradigm from binary classification to severity ranking. By coupling SFT with Rubric RLVR, KUROKO establishes an iteration-friendly workflow: its structured CoT outputs provide the diagnostic transparency needed for targeted policy refinement. For industrial scalability, the hierarchical coarse-to-fine architecture ensures cost-efficient performance at million-content scale. Despite its compact 4B size, KUROKO consistently outperforms leading proprietary VLMs, delivering significant gains in both high-risk interception and mid-to-low-tier discrimination during production A/B tests. This work offers a transferable blueprint adaptable to diverse content governance scenarios.

Limitations

While KUROKO establishes a robust foundation for severity-aware moderation, several limitations remain. First, to protect production integrity and prevent policy circumvention, certain operational details—including specific rule weights and the precise formulations of our severity aggregation functions—are redacted. Disclosing these mechanics risks exposing proprietary commercial enforcement standards. Second, while our evaluation demonstrates clear advantages over leading commercial VLMs and foundational ordinal classification baselines, future studies could incorporate broader listwise/pairwise ranking and calibrated regression frameworks to further isolate the specific drivers of our performance gains. Additionally, exploring deeper theoretical connections between our SACRM framework and other emerging verifiable RL paradigms remains a promising direction. Lastly, to achieve massive industrial scalability, our data engine utilizes cross-model verification; supplementing this automated pipeline with periodic human-in-the-loop calibration in the future could effectively mitigate the inherent priors typical of foundational VLMs.

Ethics Statement

Data Privacy and Controlled Access All content passes platform compliance review and mandatory screening excluding any depiction of potential minors before sampling—a corpus-eligibility criterion applied across all categories and tiers, independently of severity. De-identification is *task-preserving* and precedes any transfer: faces and identifying logos are masked, user metadata and account IDs stripped. Two annotators independently labelled each candidate before and after masking, and only samples with identical labels across both rounds enter the benchmark. Masking reduces direct identifiability but does not anonymize: body, setting and other contextual cues remain, and are themselves the evidence the severity label rests on. It follows the platform's open-data release guidelines and is reviewed by its Legal and Compliance department, with no material transferred before that review completes. Access is **application-based, never open download**, under signed agreements restricted to approved research use, prohibiting redistribution, re-identification and biometric use. Safe and Low-risk content supports community reproduction, whereas Medium- and High-risk tiers require stricter approval; risk tier itself does not constitute release authorization. Any item later flagged as possibly depicting a minor is withdrawn on receipt, propagating to all recipients. Code, prompts and schemas, the de-identified rule IDs, the public proxy aggregation, the SACRM implementation and aggregate statistics can be released separately without exposing raw production content or policy weights.

Annotation Quality and Human Welfare All annotators were explicitly informed about the presence of NSFW content and provided written consent. To mitigate annotation bias, labels come from our professional moderation system with cross-annotation and two-round quality control. Annotators are senior professional moderators, not crowdworkers; the workflow provides task rotation, scheduled rest, and mechanisms to pause or withdraw from exposure to sensitive material, with psychological counselling available. Approved researchers receive prominent sensitive-content warnings and handling guidance.

Cultural Subjectivity and Regulatory Grounding Severity perception varies across cultural and regional contexts. Our tiers are grounded in Chinese regulatory requirements (e.g., Cybersecurity Law, Minors Protection Regulation, Content Ecosystem Provisions). The *framework* is policy-configurable, but we do not claim these tier definitions or thresholds are culturally universal: deployment in another jurisdiction requires local legal and cultural experts to redefine the rule ontology, weights and action mapping, and to repeat human-agreement and deployment validation. The dataset card asks for recalibration rather than direct reuse.

Mitigating Over-Censorship Risks A core motivation of KUROKO is reducing over-censorship. Shifting from binary "all-or-nothing" enforcement to severity-aware ranking enables proportional platform actions with transparent, interpretable rationales. Our online evidence is indirect: at an equal human-review budget we observe higher human-confirmed actionable yield alongside reduced violative exposure and negative user feedback, rather than a directly measured reduction in over-blocking, since appeal and overturn rates were unavailable to us.

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A Details on Severity-Aware Data Engine and Dataset

To internalize the human-aligned severity reasoning introduced in Sec. 1, the model requires more than categorical labels; it necessitates a structured trajectory that mirrors expert judgment. This process involves two core challenges: identifying which policy rules are triggered and assessing overall violation severity. While identification requires the model to parse subtle contextual cues to determine which specific policy is triggered, it remains a purely qualitative step. However, rule identification alone does not equate to a final judgment; the subsequent task of severity aggregation remains a major challenge. It involves integrating multiple violation signals into a continuous risk estimate. Learning this mapping directly from human-labeled rule combinations is impractical; the combinatorial explosion of rule interactions leads to exponential supervision requirements and severe generalization failures on long-tail cases.

To address these scalability bottlenecks, we introduce a severity-aware data engine that produces severity-aware supervision through a four-stage process. To ground the model in precise rule understanding, fully manual rationale writing is prohibitively expensive, while fully automated labeling with modern VLMs is unreliablethese models excel at open-ended description but struggle with precise rule-level tagging due to safety alignment constraints and limited domain adaptation. Our pipeline therefore decouples rule matching from contextual description (Figure 3): human annotators perform closed-set rule verification at standard throughput, while Gemini-3.0-Flash (Team et al., 2023) generates rich, open-domain contextual rationales. All supervision is structured in a severity-aware CoT format for stepwise learning of moderation reasoning, as illustrated in Figure A-3.

To make aggregation scalable, we replace the memorization of infinite rule combinations with a tiered severity schema. By grouping rules into three risk tiers with predefined weights, our aggregator ϕ (Equation (1)) enables schema-guided reasoning that aligns with frontline enforcement logic. Because ϕ is a separate module rather than a learned mapping, adapting it to a different policy means redefining tier membership and weights rather than retraining; Appendix C shows our conclusions hold across 11 weight schedules. Transfer to other jurisdictions or platforms remains untested.

A.1 Four-Stage Data Production Pipeline

Our pipeline processes raw data through the following four-stage trajectory:

Stage 1: Human-Centric Rule Matching. Annotators perform closed-set rule annotation for untagged content at the same throughput as conventional workflows, ensuring no regression in annotation efficiency.

Stage 2: Generative Rationale Synthesis. Each labeled sample—whether human-annotated in Stage 1 or pre-tagged from legacy data—is forwarded to Gemini-3.0-Flash to generate fine-grained contextual descriptions explaining the violation rationale.

Stage 3: Cross-Model Logic Validation. Descriptions are validated via consensus among three independent VLMs (Gemini-3.0 Pro, QwenVL-Max, GPT-5.2). Only unanimous agreements are accepted, minimizing hallucination risks (Wang et al., 2025a).

Stage 4: Task-Specific Dataset Generation. Validated reasoning data is transformed via structured templates into three task-specific training sets: **(i)** captioning templates for fine-grained vi-

olation description, **(ii)** VQA templates for rule-grounded visual reasoning, and **(iii)** severity ranking templates implementing the aggregator ϕ for score generation.

A.2 Data Engine Annotation Efficiency

We evaluate the throughput and accuracy of fully manual annotation against our hybrid data engine. As demonstrated in Table A-1, the hybrid approach achieves a **5.6**$\times$ speedup while simultaneously improving empirical accuracy, validating the scalability advantages of decoupling rule verification from descriptive reasoning.

Human annotators exhibit significantly lower throughput on open-ended descriptive rationales compared to closed-set rule matching, as manual description writing requires substantial cognitive effort for lexical selection and contextual synthesis. Our pipeline exploits this asymmetry by assigning closed-set verification to humans and descriptive generation to VLMs, which effectively eliminates individual biases in phrasing and subjective interpretation.

Method	Time / 1K (Days)	Acc. (%)	Speedup
Fully Manual	3.5	93.5	1.0$\times$
Hybrid (Ours)	**0.6**	**95.6**	**5.6**$\times$

Table A-1: Annotation throughput and accuracy comparison.

A.3 FG-CMB Dataset Taxonomy and Distribution

The FG-CMB dataset is structured around a rigorous two-level taxonomy aligned with industrial moderation standards. It comprises 50 fine-grained violation elements (L2)—serving as specific reasoning anchorsnested within five major risk domains (L1) plus a safe class.

This hierarchical composition is visualized via a Sunburst Chart in Figure A-1, where the inner and outer rings correspond to L1 and L2 categories, respectively. As indicated by the sector proportions, the dataset preserves the natural long-tail distribution of real-world risks, ensuring that the benchmark faithfully reflects the complexity of actual industrial moderation scenarios.

B Prompt Templates

The prompt templates utilized for the hybrid data annotation pipeline are illustrated in Figure 3 of the main text. In this section, we focus on the specific instructions used for model training, inference and API evaluation. It should be noted that while the logical structure of the prompt is fully preserved to demonstrate our reasoning framework, specific moderation rules and scoring algorithms have been redacted due to commercial confidentiality.

To reconcile the conflict between the high computational cost of CoT and the strict low-latency requirements of online industrial deployment, we introduce an Adaptive CoT mechanism via a meta-cognitive directive. This allows the model to dynamically assess input ambiguity and bypass the computationally expensive reasoning phase for clear-cut cases. Specifically, the following conditional instruction is appended: *"Please evaluate the complexity of the content and autonomously decide whether to execute the full reasoning process or provide a direct assessment."*

C Reproducible Proxy Weight Schedule

The exact production weights d_p and the action thresholds they feed are replaceable *deployment-policy parameters*, not learned parameters and not an indispensable secret of the ranking method. Publishing them together with the thresholds would expose an actionable approximation of the deployed enforcement boundary: at order-10^6 daily submissions, an adaptive adversary could estimate the marginal contribution of each violation signal and assemble content that stays just below the review threshold. We therefore separate *scientific auditability* from *policy disclosure*.

For every number reported under the four-way protocol we use an executable **public proxy** aggregation, derived naively from per-rule IDs and reported here in full so that results can be recomputed without touching our production configuration. Writing H, M and L for the number of matched high-, medium- and low-risk rules:

$$
\begin{aligned}
s_{\text{High}} &= 70 + \min(30,\ 8H + 2M + L) \\
s_{\text{Medium}} &= 30 + \min(40,\ 5M + 2L) \\
s_{\text{Low}} &= \min(30,\ 3L)
\end{aligned}
$$

with counts for virtual (non-photographic) subjects scaled by 0.7. We release this scorer, the de-identified rule IDs matched by KUROKO on approved benchmark samples, the SACRM implementation and an SOP with scripts that regener-

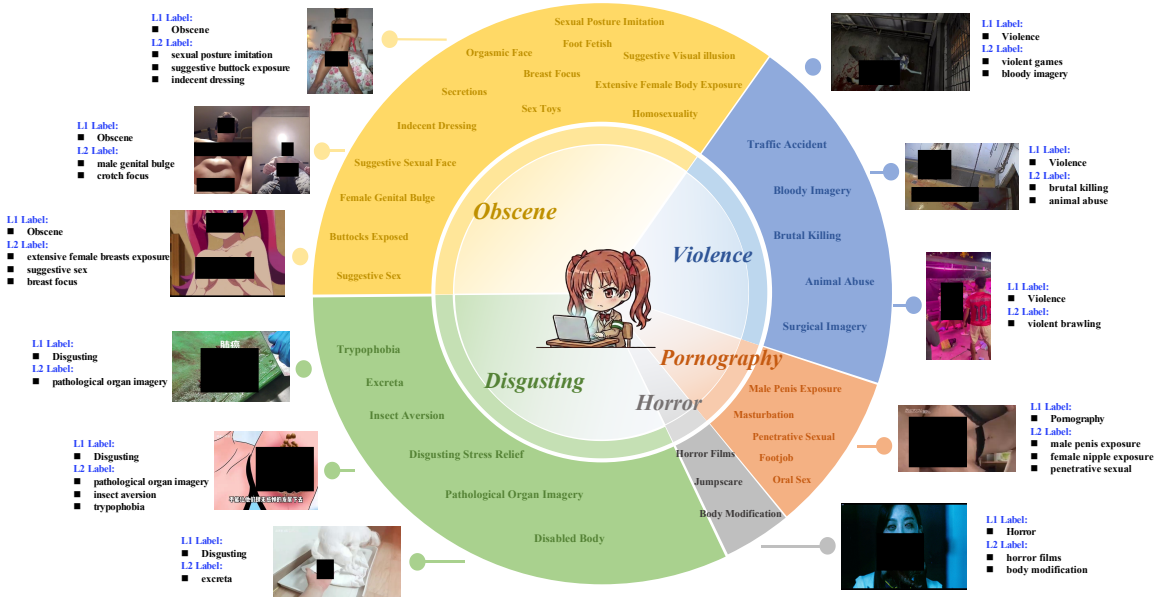


Figure A-1: FG-CMB benchmark overview. FG-CMB provides hierarchical severity annotations across five risk domains (Pornography, Obscene, Violence, Horror, Disgusting), stratified into four tiers: Safe, Low-Risk, Medium-Risk, and High-Risk.

ate every reported metric, so reproduction never requires our aggregation weights.

To confirm that the conclusion does not depend on one chosen weighting, we repeated the matched comparison under 11 proxy variants: independently perturbing the High/Medium/Low contributions by $\pm 20\%$, jointly scaling all contributions by $\pm 20\%$, and moving the virtual-content coefficient to 0.5 and 0.9. KUROKO exceeds the matched ordinal baseline in **11 of 11** variants on Spearman, Kendall, intra-tier pairwise accuracy and Harm Recall@10%; across variants Spearman stays in .6075–.6077 versus .5367–.5372 and intra-tier accuracy in 66.18–69.61% versus 57.14–58.85%. The public proxy does not reproduce the platform's exact scores or enforcement boundary; it makes rule matching, reward computation and the robustness of our central conclusion independently auditable.

D Unified-Rubric API Re-evaluation

Tables 2 and 3 give each API the same rules and an equivalent instruction, but ask only for a discrete label—the one intended difference, since a discrete label instead of a score is the contrast under study. Those API prompts admit only a violation *type*, so their per-tier columns measure stratified binary detection and involve no implicit tier mapping. To remove any doubt that the mapping itself depressed API performance, we reran the evaluation with an identical rubric and identical tier definitions, each model emitting its own tier (Table A-2).

Model	High	Medium	Low
Qwen-VL-Max	22.7/86.6/36.0	32.9/88.3/48.0	15.8/93.9/27.0
Gem.-3.0-Flash	23.8/95.3/38.1	34.2/97.0/50.6	15.9/98.1/27.4
Gem.-3.0-Pro	38.5/82.6/52.5	52.3/88.2/65.6	28.0/87.4/42.4
KUROKO-RL	**42.3**/70.6/**52.9**	**57.6**/80.1/**67.0**	**34.3**/85.6/**49.0**

Table A-2: Unified-rubric re-evaluation: identical rubric and tier definitions, each model emitting its own tier. Cells are P/R/F1 in %. GPT-5.2 is omitted because our provider withdrew access before this rerun; its rows in Tables 2–3 remain curated-prompt and we make no unified-rubric claim for it.

The two weaker APIs still sit at 15.8–34.2% precision with near-ceiling recall, so their over-flagging is not an artefact of our prompt. Gemini-3.0-Pro nearly closes the High tier (52.5 vs. 52.9 F1), while KUROKO retains its largest margin on the Low tier this paper targets (+6.5 pp). This experiment controls prompt and mapping; it does not equalise the difference between a zero-shot API and a domain-adapted model, which is why Sec. 4.3 rather than this table carries the paradigm claim.

Instruction for Severity Ranking

Input: Please analyze the image for violation severity based on the moderation rules. Strictly follow the process below:

Moderation Rules

- **High-Severity Rule Set:** A comprehensive collection of explicit violation criteria.
- **Mid-Severity Rule Set:** Numerous context-aware and compositional violation rules.
- **Low-Severity Rule Set:** Extensive boundary-sensitive rules designed for ambiguity-heavy cases.

Severity Criteria

- **High Risk (70-100):** Contains explicit high-risk elements.
- **Medium Risk (30-69):** Contains implicit or moderate violation elements without high-risk features.
- **Low Risk (1-29):** Contains only slight boundary elements
- **Safe (0):** No violations.

Score Calculation Logic:

[Detailed arithmetic formulas are redacted for proprietary reasons.]

Chain-of-Thought (CoT) Requirements Please output the analysis in the following structured format:

1. **Content Recognition:** Briefly describe potential violating content.
2. **Element Identification:** List all violation elements by risk level.
3. **Calculation Process:** State related information and show calculation.
4. **Final Scoring:** Output final score and category.

Output Format:

```
<think> [Insert full reasoning and calculation process here] </think>
<answer> [Final Answer] </answer>
```

(a) Instruction for Severity Ranking.

Instruction for Violation Type Classification

Input: Please analyze the image according to the platform moderation rules and determine whether it contains policy violations. Strictly follow the process below:

Moderation Rules

- **High-Severity Rule Set:** A comprehensive collection of explicit violation criteria.
- **Mid-Severity Rule Set:** Numerous context-aware and compositional violation rules.
- **Low-Severity Rule Set:** Extensive boundary-sensitive rules designed for ambiguity-heavy cases.

Chain-of-Thought (CoT) Requirements Please structure the analysis as follows:

1. **Risk Identification:** Categorize detected elements into corresponding rules.
2. **Decision Process:** Provide a structured explanation of why the content constitutes a violation.
3. **Final Classification:** Output the final violation type.

Output Format:

```
<think> [Insert structured reasoning here] </think>
<answer> [Final Classification] </answer>
```

(b) Instruction for Violation Type Classification.

Figure A-2: **Prompt Templates.**

E First-Stage Operating Point and Risk Governance

At the fixed production operating point, the first-stage Recall Classifier attains 98.61% recall over all violating samples. Here we document the cost structure that sets that operating point. The platform handles order-10^6 submissions daily under continuous adversarial pressure and runs two tracks: review-before-publish and publish-before-review. KUROKO serves the former, so items it clears remain subject to post-publication review, user reports and rapid re-review. Chasing the last long-tail cases at the first stage would route a large share of traffic to the VLM for a small residual that the later tracks catch more cheaply; the threshold is therefore a cost-constrained choice that we defend by measurement rather than by assuming the residual is negligible.

The harder constraint is measurement. The rates above are *benchmark* FNR on a hard-case-enriched set. Online FNR requires human re-review of a sample of the millions of items cleared daily, which is unaffordable per release; we paid for it once at small scale, and annotating a 5% sample of online traffic put the first stage's online FNR at 0.21%. Every later version is then gated by safety-coverage parity against that baseline arm—a candidate ships only if coverage is at least on par—which holds the accepted level without repeating the audit.

F Case Study

In this section, the application performance of KUROKO in content moderation and severity-aware ranking is demonstrated with cases of content published by real users. The model extracts violation elements from input visual information and generates audit conclusions by matching relevant rules. This not only manifests the excellent comprehension capability of the audit model, but also allows readers to intuitively and quickly grasp the matching logic of audit rules and the calculation method of risk scores. The case results show that KUROKO can effectively assist platforms in identifying inappropriate content and safeguard users healthy experience.

G Lightweight Auxiliary Networks

To ensure reproducibility, we detail the architectures of the two lightweight components used in KUROKO.

G.1 Recall Classifier

To balance the real-time performance and recall rate of the content moderation model, a lightweight pre-recall sub-model classifier model is deployed before KUROKO, constructing a two-stage moderation architecture of pre-recall and multimodal model fine judgment. Built on the lightweight EfficientNet backbone, this sub-model extracts and recognizes multimodal violation features in a coarse-grained way. It only supports violation/non-violation binary classification and rough high-risk violation type judgment. Non-violating samples are directly approved, while boundary-ambiguous samples are recalled to KUROKO for fine-grained analysis, precise violation rules matching and severity ranking. This pre-recall mechanism can drastically filter out safe content and significantly reduce the inference load of KUROKO model, thus improving the end-to-end moderation efficiency while ensuring a high recall rate for the entire moderation pipeline. Meanwhile, in response to the rough judgment errors of the pre-recall sub-model, dynamic recall thresholds are set for regulation, which can be flexibly adjusted according to the accuracy-efficiency requirements of practical moderation scenarios. This achieves a dynamic balance between recall rate and moderation speed, and meets the real-time screening requirements for massive multimodal content.

G.2 Complexity Router

To achieve lightweight yet effective routing, the Complexity Router is designed as a cross-attention-based module followed by a MLP classifier. Specifically, the router takes the visual features $\mathbf{V}$ from the visual encoder and the textual embeddings $\mathbf{T}$ of the instruction as input. A cross-attention layer fuses these modalities to obtain a query-aware visual representation, which is then projected by the MLP to a binary scalar $p \in [0, 1]$, representing the probability that the input requires complex reasoning.

Training this router requires ground-truth labels distinguishing "Easy" samples (Direct Mode sufficient) from "Hard" ones (CoT needed). To generate these labels, we first fine-tune the backbone model on a subset of the FG-CMB dataset using *only* complete Chain-of-Thought annotations. This ensures the pilot model possesses the necessary domain knowledge and reasoning capabilities.

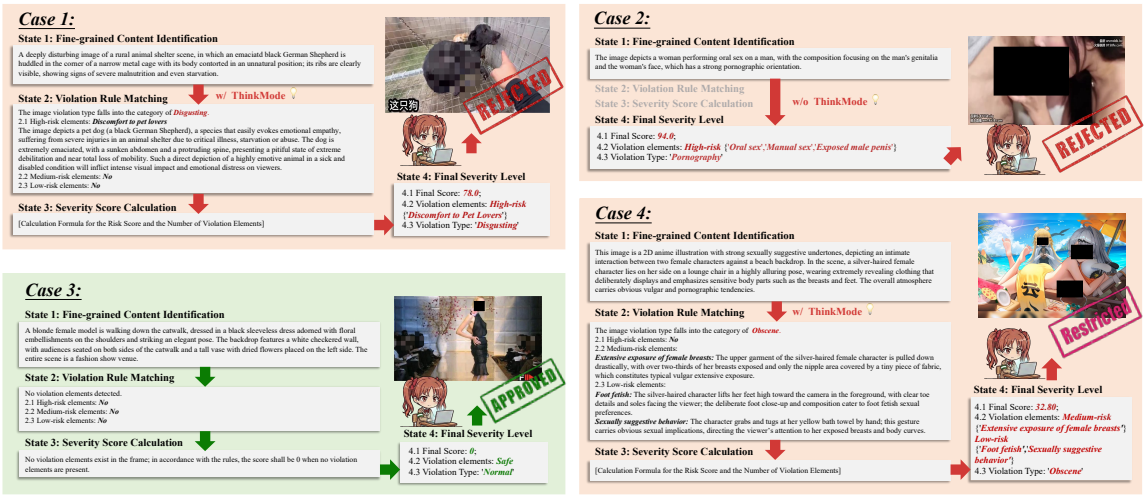


Figure A-3: Cases of KUROKO

We then employ a consistency-based difficulty assessment strategy:

1. **Sample Generation:** For each query x in the training set, we perform inference with the pilot model $N = 8$ times using the Direct Mode (i.e., enforcing immediate answer generation).

2. **Consistency Voting:** We calculate the accuracy of these N responses against the ground truth.

    - If the model correctly answers the query with high consistency (i.e., accuracy $\geq \tau_{acc}$, where we set $\tau_{acc} = 7/8$), the sample is deemed **Simple** (Label 0).
    - Conversely, if the model exhibits low consistency or high error rates in Direct Mode, the sample is labeled **Hard** (Label 1), necessitating the activation of the Chain-of-Thought path.

A critical challenge arose in defining the inference prompt for this evaluation. Initially, we attempted to enforce direct answer generation by fixing the prefix with empty tags `<think></think>`, inspired by Adaptthink (Zhang et al., 2025). However, we observed that this rigid suppression often led to format instability, where the model would generate reasoning steps inside the answer block (e.g., `<answer> [reasoning...] true answer </answer>`). To address this, following Autothink (Sharma, 2025), we explicitly fix the generation prefix to `<think>...</think>`. By providing this minimal, closed reasoning trace, we effectively signal to the model that the thinking phase is concluded, compelling it to transition immediately to the answer generation phase.